

2-Mark Questions & Answers — DBMS Module 4: Relational Database Design

Q1. Define 1NF, 2NF, 3NF, and BCNF.

- **1NF:** A relation where all attribute values are atomic (no repeating groups/multivalued cells).
- **2NF:** In 1NF and no non-prime attribute is partially dependent on any candidate key.
- **3NF:** In 2NF and no non-prime attribute is transitively dependent on the primary key ($X \rightarrow Y$ implies X is superkey or Y is prime).
- **BCNF:** For every non-trivial functional dependency $X \rightarrow Y$, X must be a strict Super Key.

Q2. What is partial functional dependency?

A **Partial Functional Dependency** occurs when a non-prime attribute is functionally dependent on only a proper subset of a composite candidate key (e.g., in key AB , if $A \rightarrow C$, C is partially dependent).

Q3. Define transitive dependency with an example.

A **Transitive Dependency** occurs when a non-prime attribute determines another non-prime attribute through a primary key ($A \rightarrow B$ and $B \rightarrow C$, therefore $A \rightarrow C$).

- Example: $Emp_ID \rightarrow Dept_ID$ and $Dept_ID \rightarrow Dept_Name$.

Q4. What is a multivalued dependency (MVD)?

A **Multivalued Dependency (MVD)** $X \twoheadrightarrow Y$ exists when the presence of two tuples sharing attribute X implies that attribute Y is completely independent of the remaining attributes Z in the relation.

Q5. What is a lossless-join decomposition?

A decomposition of relation R into R_1 and R_2 is **lossless-join** if $R = R_1 \bowtie R_2$ (joining R_1 and R_2 produces the exact original tuples without creating spurious/fake records).